



**FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA**  
**DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE**  
**END SEMESTER EXAMINATION (SEMESTER II)**  
**(December 2024)-2022/23**  
**CSC1023/CS102-Object-oriented Programming/Programming**  
**Techniques (3 Credits)**



Answer All Questions

Time allowed: Two Hours

**1. [25 Marks]**

- a) Explain what is meant by the Java development environment. **[3 marks]**
- b) Vehicles are mainly used for the purpose of transportation. When we compare different types of vehicles, they have certain features/functionalities which are in common as well as different. Write a simple Java program structure to depict the relationship between trucks, cars and motorbikes. (Mention only the necessary methods and attributes). **[6 marks]**
- c) Mention a real-life example related to computer systems that would depict the concept of polymorphism in object-oriented programming. **[3 marks]**
- d) Compare and contrast arguments and parameters in Java using appropriate examples. **[4 marks]**
- e) Explain the purpose of having “util” package in Java. **[4 marks]**
- f) Given the following code snippet:

```
public class CircleUtil {  
    private final double TWO_PI = 2 * Math.PI;  
  
    public double perimeter(final double r) {  
        return TWO_PI * r;  
    }  
}
```

Explain how you can save the memory when running the above piece of code. Justify your answer. **[5 marks]**

**2. [25 Marks]**

- a) Given the following abstract class definition:

```
public abstract class Vehicle {  
    private int speed = 0;  
    public abstract void startEngine();  
}
```

Is it possible to create objects using the above-mentioned abstract class? If your answer is yes, write down the possible implementation that will allow an abstract class to create an object. If the answer is no, justify your answer. **[5 marks]**

- b) Write a simple Java structure to show how protected access modifier is used with respect to methods and attributes. **[4 marks]**

c)

```
interface player {  
    final int id = 10;  
    int move();  
}
```

Is the above-mentioned code legal? Justify the reason for your answer. [4 marks]

- d) Cricket and Soccer are considered to be two types of sports. By using upcasting/down-casting modify the given code so that the program will output names of the sports as "Cricket" and "Soccer". [6 marks]

```
class Sport {  
    void displayName() {  
        System.out.println("Sport");  
    }  
}
```

- e) What is the sandbox in Java virtual machine? [3 marks]

- f) Is it possible to implement multiple interfaces in a single class? Justify your answer. [3 marks]

### 3. [25 Marks]

- a) Using a suitable example, explain how the abstraction is used in the Java Collections Framework. [3 marks]

- b) What will be the output of the below code? Provide reasoning for your answer. [4 marks]

```
class Parent {  
    public static final String message = "Parent Message";  
    public static void printMessage() {  
        System.out.println(message);  
    }  
}  
  
class Child extends Parent {  
    public static final String message = "Child Message";  
    public static void printMessage() {  
        System.out.println(message);  
    }  
}  
  
public class TestInheritance {  
    public static void main(String[] args) {  
        Parent p = new Child();  
        System.out.println(p.message);  
        p.printMessage();  
    }  
}
```

- c) i) Explain the String Pool concept in Java highlighting its key characteristics. [3 marks]

- ii) Analyze the following code segment and determine the output. [3 marks]

```
String str1 = "Coding";
String str2 = "Challenge";
StringBuilder sb = new StringBuilder("Try ");

str1 = str1.concat(" Fun").toUpperCase();
str2 = str2.replace("l", "L");
sb.append(str1.substring(0, 6));
sb.insert(4, str2.substring(4, 8));
sb.delete(0, 3);
System.out.println(sb);
```

- d) If a class is already marked final, is it necessary to also declare its methods as final? Justify your answer. [3 marks]
- e) Rewrite the Java class below to ensure that only a single instance of the class exists in the system at any given time during execution. [5 marks]

```
class AccessController {
    private int noOfAccounts;
    public AccessController(int noOfAccounts) {
        this.noOfAccounts = noOfAccounts;
    }
    // Other methods
}
```

- f) Determine whether the following code segments are legal or not. [4 marks (2x2)]

i)

```
public class Foo4 {
    static final int x = 12;
    public void go() {
        System.out.println(x);
    }
}
```

ii)

```
class MockSongs {
    public static List<String>
        getSongStrings() {
        List<String> songs = new
            ArrayList<>();
        songs.add("somersault");
        songs.add("havana");
        return songs;
    }
}
```

#### 4. [25 Marks]

- a) Consider the MovieSong and MusicPlayer classes provided below and write the required Java statements for the questions.

```
class MovieSong {
    private String title;
    private String artist;
    public MovieSong(String title, String artist) {
        this.title = title;
        this.artist = artist;
    }
    // other methods
}
```

```

class MusicPlayer<T> {
    private T song;
    public MusicPlayer(T song) {
        this.song = song;
    }
    public T getSong() {
        return song;
    }
}

```

- i) Instantiate two distinct MusicPlayer instances, configuring one to handle MovieSong objects and the other to manage String-based track names. [4 marks]
  - ii) Modify the MusicPlayer class, only to accept ClassicalSong and its subtypes. [3 marks]
- b) i) List the types of Java Exceptions. [2 marks]
- ii) Determine the Output of the program below. Explain your answer. [4 marks]

```

class PaymentException extends RuntimeException {}
class AuthorizationException extends RuntimeException {}

public class Main {
    public static void main(String[] args) {
        String transactionType = "Credit";
        try {
            System.out.println("Start Transaction");
            processTransaction(transactionType);
        } catch (PaymentException pe1) {
            System.out.println("Payment Processing Error");
        } catch (AuthorizationException ae2) {
            System.out.println("Authorization Failure");
        } finally {
            System.out.println("Transaction Log Completed");
        }
    }

    static void processTransaction(String transactionType) {
        System.out.println("Verifying Transaction");
        if ("Credit".equals(transactionType)) {
            throw new PaymentException();
        }
        if ("Debit".equals(transactionType)) {
            throw new AuthorizationException();
        }
        System.out.println("Transaction Approved");
    }
}

```

- iii) What are PaymentException and AuthorizationException Exception types? [2 marks]
  - iv) Is the following modification legal? [2 marks]  
 static void processTransaction(String type) throws RuntimeException  
 {...}
- c) i) What is a Character Stream in Java, and how are they handled? [3 marks]

- ii) Identify any issues (if any) in the program below, which is designed to write a String array to a file and read it back. If issues are found, make the necessary corrections.  
[5 marks]

```
public class Main {
    public static void handleDocument(String[] names) {
        BufferedWriter writer = new BufferedWriter(new FileWriter(
            "names.txt"));
        writer.write("Name List:");
        for (String name : names) {
            writer.write("\n" + name);
        }
        writer.close();

        BufferedReader reader = new BufferedReader(new FileReader(
            "names.txt"));
        String line;
        while((line = reader.readLine()) != null) {
            System.out.println(line);
        }
        reader.close();
    }

    public static void main(String[] args) {
        String[] names = {"John", "Carl", "Jerry"};
        handleDocument(names);
    }
}
```

—END—